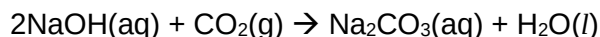


Question 1: To determine the concentration of NaOH and Na₂CO₃ in FA1

Aqueous NaOH can be used to remove CO₂ from exhaled air in spacecrafts. NaOH reacts with CO₂ according to the following equation:



A sample of aqueous NaOH was used to absorb CO₂ in the air for some time. The resulting mixture of NaOH and Na₂CO₃ formed was labelled as **FA1**.

You are to determine the concentration of NaOH and Na₂CO₃ in **FA1** by titration with an aqueous solution of HCl, **FA2**.

You are provided with the following.

FA1, a mixture of NaOH and Na₂CO₃

FA2, 2.50 mol dm⁻³ of hydrochloric acid, HCl

Methyl orange indicator

distilled water

(a) Titration with methyl orange as indicator

1. Pipette 25.0 cm³ of **FA1** into a 250 cm³ conical flask and add 1 – 2 drops of methyl orange indicator.
2. Titrate the contents of the flask with **FA2** from the burette until the solution turns orange.
3. Repeat the titration as many times as necessary to obtain consistent results.
4. Record the titration results using the table in part **(b)**. Ensure that the recorded results show the precision of your practical work.

(b) Recording of Results

Titration results with methyl orange

	1	2	
Final burette reading / cm ³	20.20	20.20	
Initial burette reading / cm ³	0.00	0.00	
Volume of FA2 / cm ³	20.20	20.20	
	✓	✓	

[3]

Comments:

- 2 consistent readings within $\pm 0.10 \text{ cm}^3$ should be given.
- All burette readings should be given to 2 decimal places.
- The final burette reading should be greater than the initial burette reading.

Obtain from your titration results a suitable average titre. Put ✓ below the readings used to calculate this average titre.

[1]

$$\text{Average titre} = (20.20 + 20.20)/2 = \underline{20.20 \text{ cm}^3}$$

Comments:

- Use the two consistent readings that are within $\pm 0.10 \text{ cm}^3$ to calculate the average titre.
- The working and statement for average titre must be clearly shown.
- The final answer for the average titre **MUST** be given to **2 decimal places** with units as the accuracy of a burette is 2 decimal places.

- (c) A student conducted the titration using 25.0 cm^3 of **FA1** against **FA2** using thymol blue as an indicator and obtained the following results.

Titration results with thymol blue

	1	2	3
Final burette reading / cm^3	14.50	28.60	42.75
Initial burette reading / cm^3	0.00	14.40	28.60
Volume of FA2 / cm^3	14.50	14.20	14.15
Chosen values		✓	✓

Select appropriate titration values from the table above by placing ticks below the chosen volumes of **FA2**.

[1]

Comments:

- Only the two readings within $\pm 0.10 \text{ cm}^3$ should be chosen. (i.e. 14.50 cm^3 should be rejected since it is not within $\pm 0.10 \text{ cm}^3$)
- All values should be recorded in ink and not pencil.

Use your chosen values to determine the volume of **FA2** reacted with 25.0 cm^3 of **FA1**. Show your working below.

$$\text{Average titre} = (14.20 + 14.15)/2 = \underline{14.18 \text{ cm}^3}$$

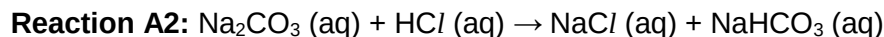
[1]

Comments:

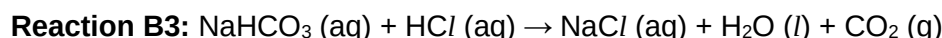
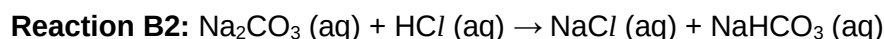
- Use the two consistent readings that are within $\pm 0.10 \text{ cm}^3$ to calculate the average titre.
- The working and statement for average titre must be clearly shown.
- A common mistake was to give the answer as 14.175 cm^3 or as 3 sig fig 14.2 cm^3 .
- The final answer for the average titre **MUST** be given to **2 decimal places** with units as the accuracy of a burette is 2 decimal places.

(d) Calculation

When **thymol blue** is used as the indicator in part **(c)**, the following reactions occur.



When **methyl orange** is used as the indicator in part **(b)**, the following reactions occur.



The NaHCO_3 in equation **B3** is the product from reaction of Na_2CO_3 with HCl in **B2**.

- (i) Calculate the volume of hydrochloric acid that reacted with 25.0 cm^3 of **FA 1** in **Reaction B3**.

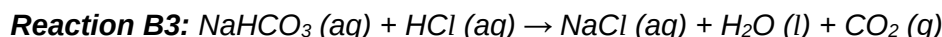
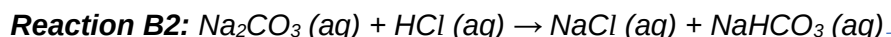


$$\begin{aligned} \text{Volume of HCl used} &= 20.20 - 14.18 \\ &= 6.02 \text{ cm}^3 \end{aligned}$$

[1]

Comments:

- Final answer should be given to 2 decimal places, following accuracy of burette.
- The volume of HCl for **reaction B1, B2 and B3** is 20.20 cm^3 (part b)
- The volume of HCl for **reaction A1, A2** is 14.18 cm^3 (part c)
- Since **Reactions A1 and A2** are identical to **Reactions B1 and B2**, we can conclude that the volume of HCl for **Reactions B1 and B2** is 14.18 cm^3 .



} 14.18 cm³

} 20.20 cm³

- Hence the volume of HCl for **Reaction B3** is the difference between the titration readings 20.20 cm^3 and 14.18 cm^3

- (ii) Considering reactions **B2** and **B3**, determine the volume of hydrochloric acid reacted with sodium carbonate in 25.0 cm^3 of **FA 1** to form NaHCO_3 .

1 mol $\text{NaHCO}_3 \equiv 1 \text{ mol } \text{Na}_2\text{CO}_3$

Volume of HCl used to react with $\text{Na}_2\text{CO}_3 = 6.02 \text{ cm}^3$

[1]

Comments:

- Final answer should be given to 2 decimal places, following accuracy of burette.
- The volume of HCl for **reaction B3** is 6.02 cm^3 found from (i).
- Since the NaHCO_3 in **Reaction B3** is a product from **reaction B2**, we can conclude that the amount of NaHCO_3 is equal to the amount of Na_2CO_3 in **reaction B2**.
- Hence, the amount of HCl required to react with NaHCO_3 in **reaction B3** is the same as the amount of HCl reacted with Na_2CO_3 in **reaction B2**.
- The volume of HCl required in **reaction B3** = **reaction B2**

Reaction B1: $\text{NaOH (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)}$

Hence, vol of HCl in B2
= 6.02 cm^3

Reaction B2: $\text{Na}_2\text{CO}_3 \text{ (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{NaHCO}_3 \text{ (aq)}$

From (i), Vol of HCl in B3 is
 $20.20 - 14.18 = 6.02 \text{ cm}^3$

Reaction B3: $\text{NaHCO}_3 \text{ (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)} + \text{CO}_2 \text{ (g)}$

14.18 cm^3
20.20 cm^3

- (iii) Calculate the volume of hydrochloric acid that reacted with the sodium hydroxide present in 25.0 cm^3 of **FA 1**.

Volume of HCl reacted with $\text{Na}_2\text{CO}_3 = 6.02 \text{ cm}^3$

Volume of HCl reacted with $\text{NaOH} = 14.18 - 6.02$
= 8.16 cm^3

[1]

Comments:

- Final answer should be given to 2 decimal places, following accuracy of burette.
- The volume of HCl for **reaction B2** is 6.02 cm^3 found from (ii).
- A total of 14.18 cm^3 of HCl reacts with **reaction B1 and B2** (recall that **reaction A1 and A2** is identical to **reaction B1 and B2**).
- Hence, the volume of HCl reacted in **reaction B1** is $14.18 \text{ cm}^3 - 6.02 \text{ cm}^3$.
- The volume of HCl required in **reaction B3** = **reaction B2**

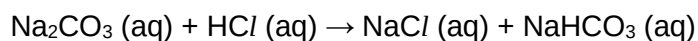
Reaction B1: $\text{NaOH (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)}$

Reaction B2: $\text{Na}_2\text{CO}_3 \text{ (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{NaHCO}_3 \text{ (aq)}$

Reaction B3: $\text{NaHCO}_3 \text{ (aq)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)} + \text{CO}_2 \text{ (g)}$

14.18 cm^3
(ii) 6.02 cm^3
20.20 cm^3

- (iv) Using your answer in (d)(ii) and the equation for **Reaction A2**, calculate the amount of sodium carbonate present in 25.0 cm^3 of **FA 1**.



Volume of HCl used to react with $\text{Na}_2\text{CO}_3 = 6.02 \text{ cm}^3$

Amount of HCl reacted with $\text{Na}_2\text{CO}_3 = 2.50 \times 6.02/1000$
 $= 1.51 \times 10^{-2} \text{ mol (to 3 s.f)}$

Since $\text{Na}_2\text{CO}_3 \equiv \text{HCl}$

Amount of $\text{Na}_2\text{CO}_3 = 1.51 \times 10^{-2} \text{ mol (to 3 s.f)}$

[1]

Comments:

- Final answer should be given to 3 significant figures with units.

(iv) Calculate the concentration of sodium carbonate, Na_2CO_3 in **FA1**.

$$[\text{Na}_2\text{CO}_3] = 1.51 \times 10^{-2} \times \frac{1000}{25}$$

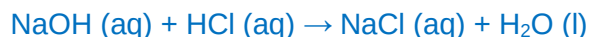
$$= 0.602 \text{ mol dm}^{-3}$$

[1]

Comments:

- Final answer should be given to 3 significant figures with units.
- The $1.51 \times 10^{-2} \text{ mol}$ of Na_2CO_3 is found in 25 cm^3 since 25 cm^3 was pipetted out and used for titration with HCl.

(v) From your answer to (d)(iii) and the equation for **Reaction A1**, calculate the amount of sodium hydroxide present in 25.0 cm^3 of **FA 1**.



Amount of HCl used $= 2.50 \times 8.16/1000$
 $= 2.04 \times 10^{-2} \text{ mol (to 3 s.f)}$

Since $\text{NaOH} \equiv \text{HCl}$

Amount of $\text{NaOH} = 2.04 \times 10^{-2} \text{ mol (to 3 s.f)}$

[1]

Comments:

- Final answer should be given to 3 significant figures with units.

(vi) Calculate the concentration of sodium hydroxide, NaOH in **FA1**.

$$[\text{NaOH}] = 2.04 \times 10^{-2} \times \frac{1000}{25}$$

$$= 0.816 \text{ mol dm}^{-3}$$

[1]

Comments:

- Final answer should be given to 3 significant figures with units.
- The 2.04×10^{-2} mol of NaOH is found in 25 cm^3 since 25 cm^3 was pipetted out and used for titration with HCl.

- (e) Give **one** reason why after 25.0 cm^3 of **FA 1** is pipetted into a conical flask, the stock solution **FA 1** should be immediately covered with a sealant film.

State the effect of not covering **FA1** with a sealant film on the calculated concentration of NaOH.

This will prevent **atmospheric carbon dioxide** from reacting further with sodium hydroxide present in **FA 1**. [1]

(accept equation)

If a stopper or sealant is not used, the calculated concentration of NaOH will be lower than actual concentration of NaOH in **FA 1** since it reacts with carbon dioxide in the atmosphere. [1]

NB: $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$

The calculated concentration of NaOH will be lower while the calculated concentration of Na_2CO_3 will be higher.

[2]

[Total: 15]**Comments:**

- Evaporation of water from the solution is negligible as compared to the NaOH absorbing CO_2 , hence the sealant is not to prevent to evaporation of water.
- NaOH is not oxidised by O_2 hence the sealant is not to prevent oxidation.
- The answer has to be clear to mention that NaOH reacts with CO_2 , not just NaOH reacts with air.

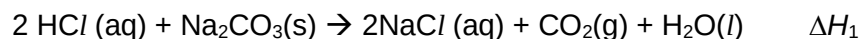
Question 2: To determine ΔH_4 , the enthalpy change of reaction between aqueous NaOH and CO₂ gas

The enthalpy change of reaction between aqueous NaOH and CO₂ gas as shown below cannot be found directly.



To determine this enthalpy change, several experiments need to be carried out.

You are to determine ΔH_1 , the enthalpy change of reaction between **FA3**, solid sodium carbonate and **FA2**, aqueous HCl.



You are provided with the following.

FA2, an aqueous solution of hydrochloric acid, HCl(aq)

FA3, solid sodium carbonate, Na₂CO₃(s)

Procedure

1. Using a 50 cm³ measuring cylinder, measure 25 cm³ of **FA2** into the styrofoam cup supported by a beaker.
2. Using a thermometer, measure and record the initial temperature (T_i) of the solution in the cup.
3. Weigh out accurately 2.50 g of **FA3** using a weighing bottle and record your weighings in an appropriate table.
4. Add the **FA3** into the styrofoam cup, and immediately cover the cup using the lid.
5. Insert the thermometer through the hole on the lid and stir the solution.
6. Measure and record the highest temperature (T_f) reached.
7. Calculate the temperature change, ΔT .
8. Reweigh the weighing bottle after emptying **FA3**. Record this mass.

(a) Table of results

Mass of empty weighing bottle /g	
Mass of weighing bottle and FA3 /g	
Mass of weighing bottle and residual FA3 /g	
Mass of FA3 used /g	2.504

[2]

Initial temperature of FA2 /°C	30.0
Final temperature/°C	36.0
Change in temperature/°C	+6.0

[2]

Comments:

- For the weighings table, the mass of **FA3** + weighing bottle (mentioned in step 3) and the mass of the weighing bottle after emptying **FA3** (or emptied weighing bottle, mentioned in step 8) are required to obtain the mass of **FA3** used.
- The mass of **FA3** must be within 2.45 to 2.55 g.
- The masses must be recorded to either 2 or 3 decimal places with units, depending on the mass balance used.
- For the temperature table, initial temperature, final temperature and the change in temperature must be recorded to 1 decimal place and units (following the accuracy of the thermometer).
- All values should be recorded in ink and not pencil.

- (b) (i) Calculate the heat evolved in the reaction between **FA3** and **FA2**.
(specific heat capacity of the mixture is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$)

$$\begin{aligned}
 q \text{ (heat released)} &= mc\Delta T \\
 &= (25)(4.18 \text{ J g}^{-1} \text{ K}^{-1})(6) \\
 &= 627 \text{ J} \\
 &= 0.627 \text{ kJ}
 \end{aligned}$$

[1]

Comments:

- The m in $mc\Delta T$ is the mass of solution. Since 25 cm^3 of solution was used and assuming density is 1 g cm^{-3} , the mass is 25g.
- Common mistakes include taking m as the mass of the solid **FA3** (2.5g) or total mass of solid and solution (25+2.5).
- This is incorrect as the solid the c is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ which is that of the solution and we are assuming that the c of the solution is the same as that of water.
- Likewise, the change in temperature is also that of the solution and not **FA3**.
- Since the units for c is $\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$, the heat change calculated will be in J.

- (ii) Determine the amount of Na_2CO_3 used.

$$\text{Amount of } \text{Na}_2\text{CO}_3 = 2.504/106 = 0.0236 \text{ mol}$$

[1]

Comments:

- The answer should be given in 3 significant figures with units

- (iii) Hence, calculate ΔH_1 , the enthalpy change of reaction between **FA3** and **FA2**.

$$\begin{aligned}
 \text{Enthalpy change of reaction} &= -0.627/0.02362 \\
 &= -26.5 \text{ kJ mol}^{-1}
 \end{aligned}$$

[1]

Comments:

The answer should be given in 3 significant figures with units.

(c) **Determination of ΔH_2 , the enthalpy change of reaction between aqueous HCl and aqueous NaOH.**

(This experiment has been conducted for you)



Procedure

1. Using the 50 cm³ measuring cylinder, transfer 25 cm³ of 2.5 mol dm⁻³ HCl into the polystyrene cup supported in a beaker.
2. Stir and record the temperature of HCl every half minute for 2.5 minutes.
3. At exactly 3 minutes, add 25 cm³ of 2.0 mol dm⁻³ NaOH using a measuring cylinder to the HCl in the cup.
4. Stir gently with the thermometer and continue to record the temperature every half minute from 3.5 minutes to 10 minutes.

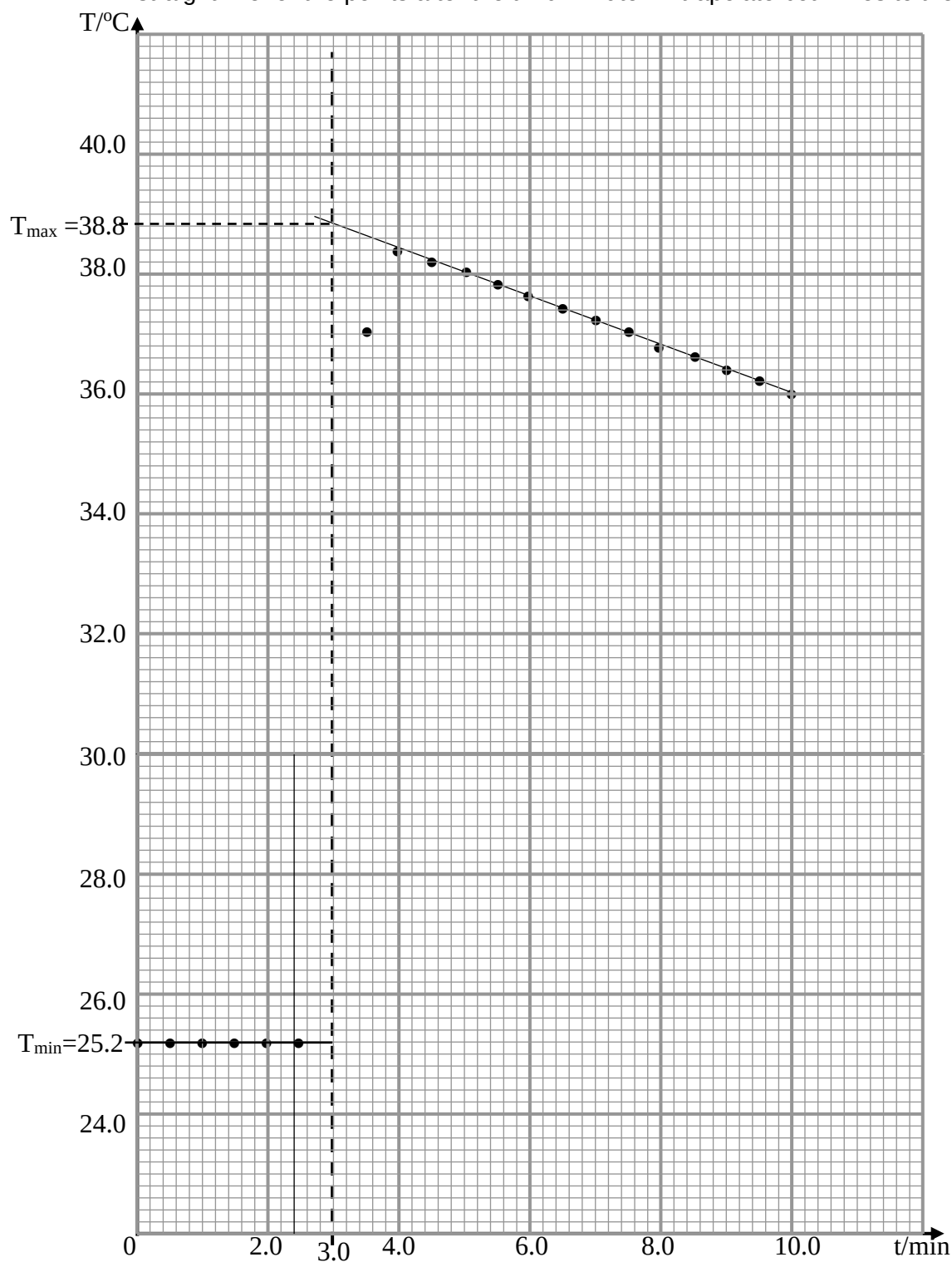
The results from the experiment are given below.

Time/ min	Temperature/ °C	Time/ min	Temperature/ °C
0.0	25.2	5.5	37.8
0.5	25.2	6.0	37.6
1.0	25.2	6.5	37.4
1.5	25.2	7.0	37.2
2.0	25.2	7.5	37.0
2.5	25.2	8.0	36.8
3.0	-----	8.5	36.6
3.5	37.0	9.0	36.4
4.0	38.4	9.5	36.2
4.5	38.2	10.0	36.0
5.0	38.0		

Plot the graph of temperature against time using the data provided above.

Use a scale of 1 large square to 2 min for the time axis and 1 large square to 2 °C for the temperature axis. The temperature axis need not start from zero.

Draw a best fit straight line for the points before the third minute. Draw a second best fit straight line for the points after the third minute. Extrapolate both lines to the third minute.



[2]

Comments:

- Draw two best fit lines from 0 to 2.5min and from 3.5 to 10min. The point at 3.5min should be ignored since it does not fit on the best fit line.
- Extrapolate both lines to 3.0min.

- (d) From your graph, read the minimum temperature, $T_{\min}$, and the maximum temperature, $T_{\max}$, at the third minute. Indicate these values clearly on your graph. Record these values in the space below and determine a value for the maximum temperature rise, ΔT , at the third minute.

$T_{\max} = \dots\dots\dots 38.8\text{ }^{\circ}\text{C} \dots\dots\dots$ (38.7 to 38.9)

$T_{\min} = \dots\dots 25.2\text{ }^{\circ}\text{C} \dots\dots\dots$

$\Delta T = \dots\dots\dots 13.6\text{ }^{\circ}\text{C} \dots\dots\dots$

[2]

Comments:

- Extrapolate the 2nd best fit line to 3.0min to obtain $T_{\max}$.
- All temperature readings should be to 1 decimal place with units.

- (e) (i) Calculate the heat evolved in the reaction between HCl and NaOH.

$$\begin{aligned} q \text{ (heat released)} &= mc\Delta T \\ &= (50)(4.18\text{ J g}^{-1}\text{ K}^{-1})(13.6) \\ &= 2842\text{ J} \\ &= 2.84\text{ kJ} \end{aligned}$$

[1]

Comments:

- The m in $mc\Delta T$ is the mass of solution. Since 25 cm^3 of HCl reacts with 25 cm^3 of NaOH at the maximum change in temperature, the volume of solution was used is 50 cm^3 and assuming density is 1 g cm^{-3} , the mass is 50g.
- Common mistakes include using only 25 cm^3 . However, to produce a temperature change of 13.6°C , the total volume is $25+25 = 50\text{ cm}^3$.
- Since the units for c is $\text{J g}^{-1}\text{ }^{\circ}\text{C}^{-1}$, the heat change calculated will be in J.

- (ii) Hence, calculate ΔH_2 , the enthalpy change of reaction between HCl and NaOH.

$$\text{Amount of NaOH (limiting)} = 25.0/1000 \times 2.0 = 0.050\text{ mol}$$

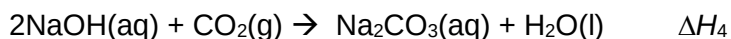
$$\begin{aligned} \text{Enthalpy change of reaction} &= -2.842/0.050 \\ &= -56.8\text{ kJ mol}^{-1} \end{aligned}$$

[2]

Comments:

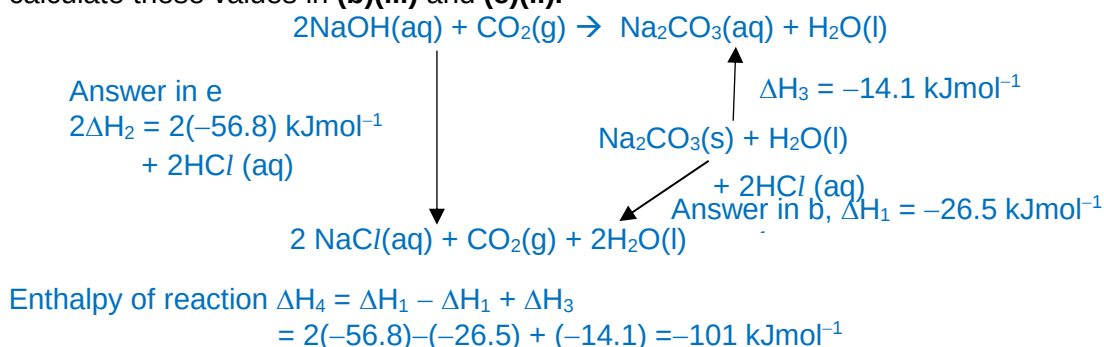
- The limiting reagent is NaOH, thus the NaOH will determine the amount of water formed.
- Since amount of NaOH is 0.050mol, the amount of water generated is 0.050mol and the enthalpy change of neutralisation will be the heat change/amt of water formed.
- Answers should be in 3 significant figures with units.

- (f) Aqueous NaOH is often used to remove CO₂ as from the air as shown below.



Given that ΔH_3 , the enthalpy of solution of Na₂CO₃(s) in water is -14.1 kJmol^{-1} , use your answer for ΔH_1 and ΔH_2 to construct an energy cycle to determine ΔH_4 .

You may use ΔH_1 as -26.5 kJmol^{-1} and ΔH_2 as -56.8 kJmol^{-1} if you did not manage to calculate these values in (b)(iii) and (e)(ii).



[4]

Comments:

- There are 2 moles of NaOH reacting with HCl, thus ΔH_2 is multiplied by 2.
- Many cycles were not balanced, forgetting to balance with HCl(aq) and correct number of H₂O(l) were a common mistakes.

[Total: 18]

Question 3: Planning

Planning 1: Determination of the percentage of sodium carbonate in FA4.

- (a) **FA4** contains a solid mixture of sodium carbonate and sodium hydroxide. You are to plan a gas collection experiment to determine the percentage by mass of sodium carbonate in **FA4**.

You may assume that the gas collected is at room temperature and pressure.

[Ar: Na: 23.0, C: 12.0, O: 16.0]

You are provided with the following:

- **FA2**: 100 cm³ of an aqueous solution of HCl.
 - Solid **FA4**
 - A 100 cm³ graduated gas syringe
 - the equipment normally found in a school or college laboratory
- (i) Calculate an appropriate mass of **FA4** which you would use for the reaction by considering the capacity of the gas syringe and assuming that **FA4** is purely Na₂CO₃.

$$\begin{aligned} \text{Amount of CO}_2 &= (90 \times 10^{-3}) \div 24 \\ &= 3.75 \times 10^{-3} \text{ mol} \end{aligned}$$



$$\begin{aligned}\text{Amount of Na}_2\text{CO}_3 &= 3.75 \times 10^{-3} \times 106 \\ &= 0.398 \text{ g}\end{aligned}$$

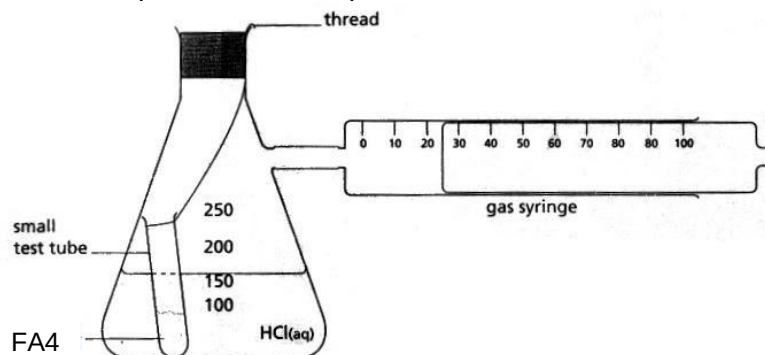
0.398 g of Na_2CO_3 will be used, assuming that **FA4** is purely Na_2CO_3 .

[2]

Comments:

- The volume of CO_2 collected can be between $50\text{-}100\text{cm}^3$. Volumes less than 50cm^3 leads to high % error while any volumes exceeding 100cm^3 will exceed the capacity of the gas syringe.
- Collection of gas over water is incorrect as it is mentioned in the question that the gas should be collected in a gas syringe.
- A common mistake was to calculate the mass of FA4 used by finding number of moles of HCl (100cm^3 of 2.5 mol dm^{-3}) and dividing by 2 to get amount of FA4. This is incorrect as the question already clearly stated to consider the capacity of the gas syringe.
- The answer should be given to 3 significant figures since this is a calculated value.

ii) Draw the experimental set-up.



[2]

Comments:

- The HCl and FA4 should be kept separate at the start to prevent gas loss. This can be done by either putting FA4 in a test tube tied with a string or adding HCl through a thistle funnel with a tap.
- A beaker should not be used to conduct the experiment as it is common to stopper a beaker.
- Likewise, a boiling tube is too small to contain 100cm^3 of HCl thus should not be used as well.
- NOTE: When drawing the gas syringe, it is important to put in the markings, i.e. the gas syringe should be graduated.

Planning 2: Determination of the water of crystallisation in FA5, hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

- (b) **FA5** contains hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$. You are to plan a gravimetry experiment to determine the value of x .

You are provided with the following:

- 1.00 g of **FA5**
- A boiling tube
- Bunsen burner
- the equipment normally found in a school or college laboratory

Outline, step by step, the procedure for the method you would use to

- carry out the weighing of a sample of **FA5** solid,
 - remove the water of crystallisation (dehydration) in **FA5** by heating, and
 - ensure that the dehydration is complete.
- Weigh accurately about 1.00g of **FA 5** in the boiling tube provided using a weighing balance.
 - Heat the tube and the contents gently first, then strongly to cause the dehydration of **FA 5** using a Bunsen Burner.
 - Place the tube on a heat-proof mat to cool, and weigh the boiling tube containing the residue.
 - Repeating the heating, cooling and weighing process until a constant mass is obtained.
 - Record all weighings clearly below in an appropriate format.

[3]

[Total: 7]

Comments:

- For the weighings, it is important to state that the FA5 is measured using a mass balance and the FA5 is weighed in a boiling tube. This is because after heating, the dehydrated FA5 will be weighed in a boiling tube so the mass of the boiling tube will be required in order to obtain the mass loss of FA5.
- The boiling tube can be placed in a beaker for weighing thus it is not necessary to weigh in a weighing bottle then transfer.
- A common mistake was not cooling the boiling tube before weighing.
- It is important to cool to room temperature before weighing so that the weighngs are done under the same conditions as before heating.
- Another common mistake was to just state that the experiment should be stopped when mass is constant but it is important to mention the repeated heat, cooling, weighing cycle till mass is constant.